Notation

The literatures this book spans grew up separately and their symbols collide; these are the conventions used here, followed by the symbols that carry more than one meaning.

PRICES, RATES, AND TIME

t	the present, or a general date
T	maturity, expiration, or settlement date
τ	accrual fraction between two dates, in years
S, S_0, S_T	spot price of the underlying: generally, now, at maturity
F, F_0	forward or futures price; a forward rate in the rates chapters
K	strike price, or the delivery price of a forward
r	continuously compounded risk-free rate
q	dividend yield
r_f	foreign risk-free rate, the yield on a currency, so $q = r_f$
y	convenience yield on a commodity
L, N	notional principal
W, dW	Wiener process and its increment

OPTIONS AND VOLATILITY

c, p	prices of a European call and put (Chapters 11, 15, 16)
C, P	prices of an American call and put (Chapter 11); P standing alone is never a bond price
σ	volatility; implied volatility where the context is a quote
$N(x)$	standard normal cumulative distribution function
$N'(x)$	standard normal density, $e^{-x^2/2}/\sqrt{2\pi}$; written φ in Chapters 29 and 36, ϕ in Chapters 24 and 25
d_1, d_2	the arguments of the Black–Scholes–Merton and Black formulas
Δ, Γ, Θ, $\mathcal{V}$	delta, gamma, theta, vega
H	barrier level
κ	speed of mean reversion of variance in a stochastic volatility model
H (Chs. 14, 21)	Hurst parameter of fractional Brownian motion, and of a rough volatility model

TERM STRUCTURE

$P(t,T)$	price at t of a zero-coupon bond maturing at T
DF_i	discount factor to date t_i (Chapters 4, 7 and 9)
$f(t,T)$	instantaneous forward rate (Chapters 31 and 32)
$A(t,T)$, $B(t,T)$	the coefficients of an affine bond price, $P = A\,e^{-Br}$
$B(t)$	value at t of one unit invested from time 0 at the overnight rate used for discounting; the accumulation account
A	annuity, or PV01 leg, of a swap
a	speed of mean reversion of the short rate
Q	the risk-neutral measure
Q^T, Q^N	the T-forward measure and the measure induced by numeraire N
E, $\mathbb{E}$	expectation, with the measure as a superscript (E^Q, E^T); the two forms are interchangeable
Λ	a Radon–Nikodym derivative: dQ/dP in Chapter 28, and the ratio of two forward measures in Chapter 30; Λ_t is its restriction to $\mathcal{F}_t$, the density process

CREDIT

λ	hazard rate, or default intensity
τ	time of default
$Q(t)$	survival probability to t (Chapters 24–26); not the measure
R	recovery rate
A	risky annuity: the present value of one basis point paid until default
ρ	default correlation

RISK

c	confidence level of VaR and ES; written q in the WCDR of Chapters 24 and 26
μ	drift, or mean
ρ	correlation
β	sensitivity to a common factor

SYMBOLS WITH MORE THAN ONE MEANING

- ρ: a *correlation* nearly everywhere (including price against variance in a stochastic volatility model), and the Greek *rho* in Chapter 18.
- H: a *CSA threshold* in Chapter 9, a *barrier level* in Chapters 34 and 35, the *Hurst parameter* in Chapters 14 and 21, a *futures price* in Chapter 28, a *haircut* in Chapter 27; no chapter uses two at once.
- λ: a *hazard rate* in Chapters 9, 24 and 25, the *market price of risk* in Chapter 28, a *jump intensity* in Chapter 21, the *EWMA decay* in Chapter 38, the *barrier exponent* in Chapter 34.
- B: the *accumulation account* $B(t)$ (Chapters 28 and 30) and an *affine coefficient* $B(t,T)$ (Chapter 31); the arguments separate them.
- τ: an *accrual fraction* in the rates chapters, the *time of default* in Chapters 9, 24 and 25, the *first call date* of an autocallable in Chapter 35.
- A: an *annuity* in the swap chapters, a *risky annuity* in the credit chapters; the same construction, stopping or not at default.
- φ, ϕ: the normal density in Chapters 24, 25, 29 and 36, Heston's *characteristic function* in Chapter 21, the call/put *sign* ± 1 in Chapter 34, a deterministic *shift* $\phi(t,T)$ in Chapter 32; Φ_k in Chapter 32 is a volatility *multiplier*, not a distribution function.

The normal distribution function is $N(\cdot)$, not $\Phi(\cdot)$, except in Chapter 36's weather options. $P(t,T)$ is always a bond price; P alone is the American put of Chapter 11, or in Chapters 19, 21 and 22 a European put as a function of strike, $P(K)$.

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